

Institute of Actuaries of India

Subject CT4 – Models

September 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) The objectives of the modelling exercise :**

- The validity of the model for the purpose to which it is to be put.
- The validity of the data to be used.
- The validity of assumptions used.
- The possible errors associated with the model or parameters used not being a perfect representation of the real world situation being modelled
- The impact of correlations between the random variables that “drive” the model.
- The extent of correlations between the various results produced from the model.
- The current relevance of models written and used in the past.
- The credibility of the data input.
- The credibility of the results output.
- The dangers of spurious accuracy.
- Cost of buying or constructing, and of running the model.
- Ease of use and availability of suitable staff to use it.
- Risk of model being used incorrectly or with wrong inputs.
- The ease with which the model and its results can be communicated.
- Compliance with the relevant regulations.
- Clear documentation.

[4]

ii) Pension scheme for medium sized client

Pension scheme for medium sized client

- Validity of data/assumptions. Compliance with legislation.
- It is a financially significant figure which you cannot afford to be way off the mark and is likely to make a big difference to the company making the contributions, so accurate data and calculations are important and compliance with legislation essential.

Cake list

- Dangers of spurious accuracy.
- The sum of money concerned is so small anything which is time-consuming or expensive is a waste.

[2]

[6 Marks]**Solution 2:****i) Strengths of Binomial model**

- The estimate is easy to calculate and avoids numerical solution of equations
- can be generalised to give the Kaplan-Meier estimate

Weaknesses of Binomial model

- The model does not use the actual times of death, so there is some loss of information

- requires calculation of initial exposed-to-risk which is unnecessarily complicated if census-type data are available, poisson model is easy to fit
- not so easily generalised for multiple decrements but two-state model can be easily generalised
- estimate of q_x has a higher variance than that of the two-state Poisson models (though the difference is very small unless mortality is very high) [2]

ii) Writing down the data in terms of a,b,d and t.

Time in year				
Life i	a	b	d	T
1	0	1	0	-
2	0	0.75	1	0.5
3	0.25	1	1	0.75

$$\text{Likelihood} = (1 - q_x)^* {}^{0.75}q_x * {}^{0.75}q_{x+0.25} \quad [2]$$

iii) Using Balducci's assumptions-

$${}^{0.75}q_x = 0.75 * q_x / (1 - 0.25 * q_x)$$

$${}^{0.75}q_{x+0.25} = 0.75 * q_x \quad (b=1, a=0.25)$$

$$\text{Therefore Likelihood} = (1 - q_x)^* {}^{0.75}q_x / (1 - 0.25 * q_x)^* {}^{0.75}q_x$$

$$L = (1 - q_x)^* {}^{0.75}q_x / (1 - 0.25 * q_x)^* {}^{0.75}q_x$$

[2]

iv) Simplifying and taking logs

$$\text{Log } L = \text{Log}(1 - q_x) + 2 \text{Log}(0.75) + 2 \text{Log}(q_x) - \text{Log}(1 - 0.25 * q_x)$$

Differentiating w.r.t q_x

$$\text{Log}'L = -1/(1 - q_x) + 2/q_x + 0.25/(1 - 0.25 * q_x)$$

$$\text{Log}'L = \{-q_x * (1 - 0.25 * q_x) + 2 * (1 - q_x) * (1 - 0.25 * q_x) + 0.25 * q_x * (1 - q_x)\} / \{q_x * (1 - q_x) * (1 - 0.25 * q_x)\} = 0 \quad (\text{setting equal to zero})$$

$$\text{Log}'L = -q_x * (1 - 0.25 * q_x) + 2 * (1 - q_x) * (1 - 0.25 * q_x) + 0.25 * q_x * (1 - q_x) = 0$$

$$-q_x + 2 - 2.5 * q_x + 0.5 * q_x^2 + 0.25 * q_x = 0$$

$$0.5 * q_x^2 - 3.25 * q_x + 2 = 0$$

$$2 * q_x^2 - 13 * q_x + 8 = 0$$

Solving for q_x gives $q_x = 0.68826$ and $q_x = 5.81174$ (reject as $q_x > 1$).

$$\text{Log}''L = d / q_x \{-1/(1 - q_x) + 2 / q_x + 0.25 / (1 - 0.25 * q_x)\}$$

$$\text{Log}''L = -1 / (1 - q_x)^2 + 2 / q_x^2 + 0.0625 / (1 - 0.25 * q_x)^2$$

Evaluating for $q_x = 0.68826$

$\text{Log}''L = -14.420$ which is less than zero.

Therefore $q_x = 0.68826$ is the maximum likelihood estimate.

Note: ($\text{Log}''L$ is greater than zero for $q_x = 5.81174$)

[3]

[9 Marks]

Solution 3:

i) With a continuous-time Markov jump process the time spent in each state is exponentially distributed. [1]

ii) The generator matrix is given below.

$$T = \begin{matrix} & \begin{matrix} A & S & T & N & E \end{matrix} \\ \begin{matrix} A \\ S \\ T \\ N \\ E \end{matrix} & \begin{bmatrix} -\frac{1}{\alpha} & \frac{1}{\alpha} & 0 & 0 & 0 \\ 0 & -\frac{1}{\beta} & \frac{\rho_2}{\beta} & \frac{\rho_3}{\beta} & \frac{\rho_1}{\beta} \\ 0 & 0 & -\frac{1}{\gamma} & 0 & \frac{1}{\gamma} \\ 0 & 0 & 0 & -\frac{1}{\delta} & \frac{1}{\delta} \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

[2]

iii) The general formula for the Kolmogorov forward differential equation in the time homogeneous case is given as:

$$\frac{dp_{ij}(t)}{dx} = \sum_k p_{ik}(t) \mu_{kj}$$

Applying this with $i = A$ and $j = S$ and removing (t) for ease.

$$\frac{dp_{AS}}{dx} = p_{AA}\mu_{AS} + p_{AS}\mu_{SS} ; \text{ As all other } \mu_{*S} \text{ are zero}$$

$$\frac{dp_{AS}}{dx} = \frac{1}{\alpha} p_{AA} - \frac{1}{\beta} p_{AS}$$

similarly, applying this with $i = A$ and $j = T$ and removing (t) for ease.

$$\frac{dp_{AT}}{dx} = p_{AS}\mu_{ST} + p_{AT}\mu_{TT}$$

$$\frac{dp_{AT}}{dx} = \frac{\rho_2}{\beta} p_{AS} - \frac{1}{\gamma} p_{AT}$$

[3]

iv) Since there are no transition from other states to A, p_{AA} is same as $\overline{p_{AA}}$

$$p_{AA} = \overline{p_{AA}} = e^{-\int_0^t \mu_{AS} ds} = e^{-\int_0^{\frac{t}{\alpha}} \frac{1}{\alpha} ds} = e^{-\frac{t}{\alpha}}$$

[2]

v) Substituting p_{AA} in equation derived in (iii)

$$\frac{dp_{AS}}{dx} = \frac{1}{\alpha} p_{AA} - \frac{1}{\beta} p_{AS}$$

$$\frac{dp_{AS}}{dx} = \frac{e^{-\frac{t}{\alpha}}}{\alpha} - \frac{1}{\beta} p_{AS}$$

$$\frac{dp_{AS}}{dx} + \frac{1}{\beta} p_{AS} = \frac{e^{-\frac{t}{\alpha}}}{\alpha}$$

Solving using integrating factor method.

$$IF = e^{\int_0^t \frac{1}{\beta} ds} = e^{\frac{t}{\beta}}$$

Multiplying both sides by IF.

$$e^{\frac{t}{\beta}} \frac{dp_{AS}}{dx} + e^{\frac{t}{\beta}} \frac{1}{\beta} p_{AS} = \frac{e^{\frac{t}{\beta}} e^{-\frac{t}{\alpha}}}{\alpha}$$

$$\frac{d(e^{\frac{t}{\beta}} p_{AS})}{dx} = \frac{e^{t(\frac{1}{\beta} - \frac{1}{\alpha})}}{\alpha}$$

Integration will give

$$e^{\frac{t}{\beta}} p_{AS} = \frac{\beta e^{t(\frac{1}{\beta} - \frac{1}{\alpha})}}{\alpha - \beta} + C$$

When $t=0$, $p_{AS} = 0$

$$0 = \frac{\beta}{\alpha - \beta} + c$$

$$C = \frac{-\beta}{\alpha - \beta}$$

Therefore:

$$e^{\frac{t}{\beta}} p_{AS} = \frac{\beta e^{t(\frac{1}{\beta} - \frac{1}{\alpha})}}{\alpha - \beta} - \frac{\beta}{\alpha - \beta}$$

$$p_{AS} = \frac{\beta e^{-\frac{t}{\alpha}}}{\alpha - \beta} - \frac{\beta e^{-\frac{t}{\beta}}}{\alpha - \beta}$$

[3]

[11 Marks]**Solution 4:**

i) Types of censoring presents:

- Type I censoring present because the study ends at a predetermined duration of 45 days.
- Type II censoring is not present because the study did not end after a predetermined number of patients had died.
- Random censoring is present because the duration at which a patient left hospital before the study ended can be considered as a random variable.
- Right Censoring is present for those lives that exit before the end of investigation period

[2]

ii) The censoring is likely to be informative.

The patients who died were probably recovering less well than patient who discharged from the hospital.

If they had not died, they would likely to remain in the hospital for longer than those who were not censored.

[1]

iii) The Kaplan-Meier estimate of the survival function is estimated as follows:

T	n	d	c	d/n	(1 - d/n)	s(t)
0	13					
5	13	1	0	0.0769	0.9231	0.92

7	12	1	0	0.0833	0.9167	0.85
14	11	1	2	0.0909	0.9091	0.77
28	8	1	2	0.1250	0.8750	0.67
35	5	1		0.2000	0.8000	0.54

So the value survival function at end of investigation period is 0.54

Assumptions:

- The censoring happens just after the death
 - Ignoring the discharge on any other ground except recovery from illness
 - Ignore any admission period before the start of investigation
- [4]

iv) Comments:

- The survival of a patient from the infection who given treatment is around 50% in light of the answer in c) above.
 - However, the hospital excluded the number of deaths who died within two weeks of observation period.
 - It also ignores the admission pre investigation period
 - It is assuming that the censored patient at the end of investigation will survive for sure.
 - Also ignoring the patients being discharged on any other ground like shifting to another hospital etc.
 - It claims that 8 out of 10 patients who responded the treatment beyond two weeks would survive.
 - So, the claims have to be viewed with respect to above considerations.
- [3]

[10 Marks]

Solution 5:

i)

- First choose a class of model which might be supposed, for physical reasons, to provide a reasonable fit to the data. Identify the parameters of the model.
 - Next estimate the values of the parameters from the data.
 - Check if the observed data match the pattern which would be expected if the model were accurate and if the parameters had the values given by their estimates. If not, the model should be revised.
- [3]

ii)

- The time spent in state H before the next visit to S has mean σ^{-1}
 - Therefore a reasonable estimate for $\hat{\sigma}$ is the reciprocal of the mean length of each visit: = (Number of transitions from H to S)/(Total time spent in state H up until the last transition from H to S), although it would be equally valid to use the Maximum Likelihood Estimator, which is (Number of transitions from H to S)/(Total time spent in state H).
 - Similarly for $\hat{\rho}$.
- [2]

iii)

Testing whether the successive holding times are independent. Exponential variables would be best, and any procedure which does test this is acceptable. Something like using the Chi-

Square goodness-of-fit test on the even-numbered holding times, then again on the odd-numbered ones.

Any other reasonable test is also ok.

[2]

[7 Marks]

Solution 6:

- i) The principle of correspondence states that the death data and the exposed to risk must be defined consistently, i.e. the numerator d_x and denominator E_x^c must correspond.

The deaths data and the exposed to risk data come from different sources, e.g. deaths data come from claims department of an office, and whereas exposed to risk data are based on policies in force, which come from a different part of the office. So, it may be difficult to ensure the principle of correspondence.

[2]

- ii) The population and death data are based on age last birthday and hence no age adjustment is needed here. Assuming population varies linearly between census dates.

$$\text{Population as on 1.1.2015} = \frac{1}{4} 1210 + \frac{3}{4} 1350 = 1315$$

$$\text{Population as on 1.1.2016} = \frac{1}{4} 1350 + \frac{3}{4} 1470 = 1440$$

The central exposed to risk for lives aged 41 last birthdays for the calendar year 2015 is given by

$$\begin{aligned} E_{41}^c &= \frac{3}{12} \times \frac{1}{2} (1315 + 1350) + \frac{9}{12} \times \frac{1}{2} (1350 + 1440) \\ &= 333.125 + 1,046.25 = 1379.375 \end{aligned}$$

The death rate is given by

$$\mu_x = \frac{\theta_{x,t}}{E_x^c} = \frac{14}{1379.375} = 0.010149$$

[3]

- iii) For each life observed during the investigation period we need following data:
- Date of birth
 - Date of joining the investigation (if after 31.03.2014)
 - Date of leaving the investigation (if before 31.03.2016)
 - Reason for leaving investigation (death or any other reason)

[2]

[7 Marks]

Solution 7:

- i) Let the states be “using brand A”, “using brand B”, and “using brand C”. Then the transition matrix T is

$$T = \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0 & 0.6 & 0.4 \\ 0.2 & 0.4 & 0.4 \end{bmatrix}$$

All the probabilities in a row adds up-to 1, therefore the transition probabilities are consistent.

[2]

- ii) The probability of customer using brand B in six weeks' time if he is currently using brand A is given by p_{12}^2 , so let's find T^2

$$T^2 = \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0 & 0.6 & 0.4 \\ 0.2 & 0.4 & 0.4 \end{bmatrix}^2 = \begin{bmatrix} 0.31 & 0.34 & 0.35 \\ 0.08 & 0.52 & 0.4 \\ 0.18 & 0.44 & 0.38 \end{bmatrix}$$

$$\text{Thus, } p_{12}^2 = .34$$

[1]

$$\text{iii) } T^3 = T * T^2$$

$$T^3 = \begin{bmatrix} 0.5 & 0.2 & 0.3 \\ 0 & 0.6 & 0.4 \\ 0.2 & 0.4 & 0.4 \end{bmatrix} \times \begin{bmatrix} 0.31 & 0.34 & 0.35 \\ 0.08 & 0.52 & 0.4 \\ 0.18 & 0.44 & 0.38 \end{bmatrix} = \begin{bmatrix} 0.225 & 0.406 & 0.369 \\ 0.12 & 0.488 & 0.392 \\ 0.166 & 0.452 & 0.382 \end{bmatrix}$$

Now, $p_0 = (.3 \ .2 \ .5)$ and, omitting the computations involved in finding $p_0 T^3$ it turns out that:

$$p_0 T^3 = (.1745 \ .4454 \ .3801).$$

So in three weeks the percentage of customers using brand C is about 38%.

[2]

- iv) The probability of customer using brand B in six weeks' time if he is currently using brand A is given by p_{12}^6 .

This can be computed by computing T^6 OR Multiplying first row with second column of T^3 .

$$T^6 = T^3 X T^3 \text{ OR Multiplying first row with second column of } T^3 .$$

$$T^6 = \begin{bmatrix} 0.1606 & 0.4563 & 0.3831 \\ 0.1506 & 0.4640 & 0.3853 \\ 0.1550 & 0.4606 & 0.3844 \end{bmatrix}$$

$$p_{12}^6 = 45.63\%$$

Or

Using T^3 matrix

$$p_{12}^6 = \begin{vmatrix} 0.225 & 0.406 & 0.369 \\ 0.406 & 0.488 & 0.452 \\ 0.452 & 0.452 & 0.452 \end{vmatrix} = 0.4563 = 45.63\%$$

[2]

- v) The process is irreducible because each state can be reached from other state. Also, all three states are aperiodic. Therefore the entire process is aperiodic. Therefore the process has long term stationary distribution.

It is given by $sT = s$, where $s = \{x, y, z\}$ is the long term distribution with proportion in each state given by x, y and z for state 1, 2 and 3 respectively.

$$\begin{vmatrix} x & y & z \end{vmatrix} \begin{vmatrix} 0.5 & 0.2 & 0.3 \\ 0 & 0.6 & 0.4 \\ 0.2 & 0.4 & 0.4 \end{vmatrix} = \begin{vmatrix} x \\ y \\ z \end{vmatrix}$$

This is equivalent to the following system of equations:

$$.5x + .2z = x$$

$$.2x + .6y + .4z = y$$

$$.3x + .4y + .4z = z$$

Also

$$x + y + z = 1 \text{ (sum of proportion of all states should be equal to 1)}$$

Solving these 4 equations.

$$X = 2/13 \text{ (15.38\%)}$$

$$Y = 6/13 \text{ (46.15\%)}$$

$$Z = 5/13 \text{ (38.46\%)}$$

[3]

[10 Marks]

Solution 8:

i) **Gompertz Law:**

Gompertz Law is an exponential function, and it is often a reasonable assumption for middle and older ages. It can be expressed as follows:

$$\lambda_x = Bc^x; \text{ where, } \lambda_x \text{ is a force of mortality at age } x.$$

[1]

ii) Substituting, $B = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$; into the Gompertz model,

$$\lambda_x = \exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2) \cdot c^x; \text{ defining } x \text{ as duration since 50}^{\text{th}} \text{ birthday.}$$

The hazard can therefore be factorized into two parts:

$\exp(\beta_0 + \beta_1 X_1 + \beta_2 X_2)$, which depends only on the values of the covariates, and c^x , which depends only on duration.

So, the ratio of between the hazards for any two persons with different characteristics does not depend on duration, and so the model is a proportional hazards model. [3]

iii) The baseline hazard in this model relates to a non-smoker female. [1]

iv) For a female cigarette smoker, we have

$$X_1 = 0 \text{ and } X_2 = 1 \text{ and } x = 4$$

Therefore the hazard at age 54 is given by

$$\lambda_x = \exp(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1) \cdot c^4$$

$$= \exp(-4 + 0.65) \times 1.05^4$$

$$= 0.0351 \times 1.2155$$

$$= 0.04266$$

[2]

v) The hazard for a non-smoker at duration, 's' is given by the formula

$$\lambda_s = \exp(\beta_0 + \beta_1 X_1) \cdot c^s$$

The hazard for a smoker at duration, 't' is given by the formula

$$\lambda_t^* = \exp(\beta_0 + \beta_1 X_1 + 0.65) \cdot c^t$$

If the smoker's and non-smoker's hazards are the same, then

$$\lambda_s = \lambda_t^*$$

$$\text{i.e., } \exp(\beta_0 + \beta_1 X_1) \cdot c^s = \exp(\beta_0 + \beta_1 X_1 + 0.65) \cdot c^t$$

$$\text{i.e., } c^s = \exp(0.65) \cdot c^t$$

$$\text{i.e., } c^{s-t} = \exp(0.65) = 1.9155$$

Since, $c = 1.05$

$$\text{Hence, } 1.05^{s-t} = 1.9155$$

$$\text{So, } s-t = \ln(1.9155)/\ln(1.05) = 0.65/0.04879$$

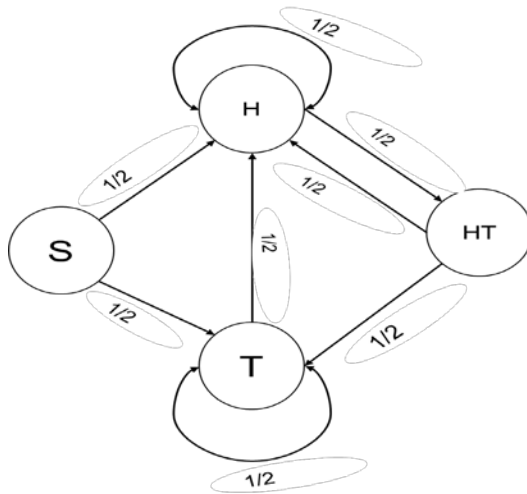
$$s-t = 13.32$$

Hence, when the two hazards are equal, the non-smoker is approximately 13 years older than the smoker. [4]

[11 Marks]

Solution 9:

i)



[2]

ii)

S	H	T	HT
0	0.5	0.5	0
0	0.5	0	0.5
0	0.5	0.5	0
0	0.5	0.5	0

This being a Markov process, the information upto 3rd toss is redundant.

The probability of head in 4th toss is $\frac{1}{2}$. Similarly probability of tail in 5th toss is $\frac{1}{2}$.

Therefore probability of head in fourth and tail in 5th toss is given by $\frac{1}{2} * \frac{1}{2} = \frac{1}{4}$

[2]

iii) The chain is finite (4 states), not irreducible (state S cannot be reached from any other state). However State S is just initial state and can be discarded after the first toss.

States H, T and HT are Irreducible (as all states can be reached from any other state). Also states H and T are aperiodic.

Therefore the process will have long term stationary distribution

[2]

- iv) Let t_i be the expected time until the state HT is reached, starting in state i, i.e., the mean first passage time to reach state HT starting in state i. Note that t_S is the expected number of tosses until first observing heads directly followed by tails. We have,

$$t_S = 1 + \frac{t_H}{2} + \frac{t_T}{2}$$

$$t_T = 1 + \frac{t_H}{2} + \frac{t_T}{2}$$

$$t_H = 1 + \frac{t_H}{2}$$

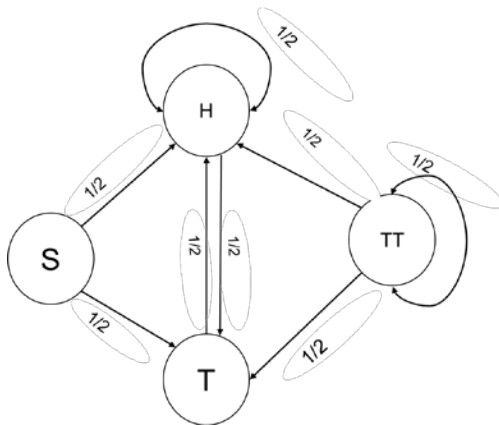
Solving these equations we get $t_H = 2$, $t_T = 4$ and $t_S = 4$, The expected number of tosses until first observing heads directly followed by tails is $t_S = 4$ [2]

- v) To find the expected number of additional tosses necessary to again observe heads followed by tails, we recognize that this is the mean recurrence time t_{HT}^* of state HT. This can be determined as

$$\begin{aligned} t_{HT}^* &= 1 + p_{HTH} * t_H + p_{HTT} * t_T \\ &= 1 + \frac{1}{2} * 2 + \frac{1}{2} * 4 \\ &= 4 \end{aligned}$$

[2]

- vi) Let's consider a Markov chain with states S,H,T,TT, where S is a starting state, H indicates heads on the current toss, T indicates tails on the current toss(without tails on the previous toss), and TT indicates tails over the last two tosses. The transition probabilities for this Markov chain are illustrated below in the state transition diagram:



[2]

- vii) Let t_i be the expected time until the state HT is reached, starting in state i, i.e., the mean first passage time to reach state HT starting in state i. Note that t_S is the expected number of tosses until first observing heads directly followed by tails. We have,

$$t_S = 1 + \frac{t_H}{2} + \frac{t_T}{2}$$

$$t_T = 1 + \frac{t_H}{2}$$

$$t_H = 1 + \frac{t_H}{2} + \frac{t_T}{2}$$

and by solving these equations we get $t_H = 6$, $t_T = 4$ and $t_S = 6$. We find that the expected number of tosses until first observing two consecutive tails is $t_S = 6$

[1]

- viii) To find the expected number of additional tosses necessary to again observe heads followed by tails, we recognize that this is the mean recurrence time t_{HT}^* of state TT. This can be determined as

$$\begin{aligned} t_{TT}^* &= 1 + p_{TTH} * t_H + p_{TT,TT} * t_T \\ &= 1 + \frac{1}{2} * 6 + \frac{1}{2} * 4 \\ &= 6 \end{aligned}$$

[1]

- ix) It may be surprising that the average number of tosses until the first two consecutive tails are observed is greater than the average number of tosses until heads is directly followed by tails, considering that the mean recurrence time between pairs of tosses with heads directly followed by tails equals the mean recurrence time between pairs of tosses that are both tails (or equivalently, the long-term frequency of pairs of tosses with heads followed by tails equals the long-term frequency of pairs of tosses with two consecutive tails). Note that the distribution of the first passage time to reach state HT (or TT) starting in state S is the same as the conditional distribution of the recurrence time of state HT (or TT), given that it is greater than 1.

[2]

[16 Marks]

Solution 10:

i)

a. Methods of graduation:

- By parametric formula: Should be used for large experiences, especially if the aim is to produce a standard table;
- With reference to a standard table should be used if a standard table for a class of lives similar to the experience is available and the experience in which we are interested does not provide much data.
- Graphical if a quick check is needed or data are very scanty.

[3]

- ii) The null hypothesis is that the graduated rates are the same as the true underlying rates for the block of business.

The test statistic

$$\chi_n^2 = \sum_x^{all\ ages} Z_x^2$$

where n is the degrees of freedom.

Age	Exposed to risk	Observed Death	Graduated Rates (qx)	Expected Death	Zx	Zx^2
20	1060	12	0.0125	13.2500	-0.3434	0.1179
21	1250	14	0.0118	14.7500	-0.1953	0.0381
22	1210	16	0.0134	16.2140	-0.0531	0.0028
23	700	9	0.0102	7.1400	0.6961	0.4845
24	875	11	0.0111	9.7125	0.4131	0.1707
25	950	15	0.0137	13.0150	0.5502	0.3027
26	805	10	0.0126	10.1430	-0.0449	0.0020
27	1390	16	0.0127	17.6530	-0.3934	0.1548
28	1080	17	0.0122	13.1760	1.0535	1.1098
29	1310	14	0.0131	17.1610	-0.7631	0.5822

Total 0.9197 2.9657

Observed Test Statistic is 2.9657

The number of age groups (n) is 10, but we lose at least one number of degrees since we are comparing our data with a set of graduated data derived from our data, perhaps 2. So the number of degree of freedom is 8.

The critical value of the chi-squared distribution with 8 degrees of freedom at the 5% level is 15.51.

Since $2.9657 < 15.51$

We do not reject the null hypothesis.

[4]

- iii) The rating factor is equal to the ratio of actual number of death to the expected number of deaths. The rating for the given data is $= 134 / 132.21 = 1.014$
- The amendment will slightly reduce the problem of the graduated table rates being low, without changing the shape of the mortality curve.

- The chi squared value would not be zero since the test looks at the sum of the deviations. However, there will be slight increase in the chi square value.

[3]

iv) Shortcomings of chi-square test:

- Failure to detect outliers - Since all the information is summarised in one number, a few large deviations may be offset or hidden by a large number of small deviations.
- Small bias - Since the squares of the differences are used, the sign of the differences are lost, hence small but consistent bias above or below may not be noticed.
- Failure to detect Clumps or runs - Again because the squares of the differences are used, the sign of the differences are lost, so significant groups of (clumps or runs) of bias over ranges of the data may not be detected.

Test used for each of the shortcomings:

- A few large deviations or outliers – Individual Standardised Deviations Test.
- Small but consistent bias – Signs Test OR Cumulative Deviations Test.
- Clumps or runs of bias over ranges of the data - Grouping of Signs Test OR Serial Correlations Test.

[3]

[13 Marks]
